Class 17 Practice Assignment

Irreversible gas phase reaction (1) is exothermic with a heat of reaction equal to –12.9 kcal mol^{-1}. The rate expression is presented in equation (2), where the pre-exponential factor equals 1.75 x 10^9 $L^{0.5}$ $mol^{-0.5}$ s^{-1} and the activation energy is 35.7 kcal mol^{-1}. The heat capacities of A, B, Y, and Z are 37, 34, 36, and 38 cal mol^{-1} K^{-1}, respectively. If an adiabatic 5.0 L BSTR is charged with equal molar amounts of A and B at 400 °C and 1 atm, how long will it take to convert 50% of reagent A and what will the temperature equal?

$$A + B \rightarrow Y + Z \tag{1}$$

$$r_1 = k_1 C_A \sqrt{C_B} \tag{2}$$